

Maximum Torque per Ampere (MTPA) Control for IPMSM Drives Using Signal Injection and an MTPA Control Law

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Abstract—This paper presents a maximum torque per ampere (MTPA) control scheme for interior permanent magnet synchronous motor drives. In this scheme, an enhanced signal injection technique is responsible for correcting the MTPA reference deviation due to motor parameter variations, and a derived MTPA control law is responsible for generating constant-parameter model-based MTPA current references to enhance the dynamic performance of the MTPA operation. Unlike previous signal injection techniques, the enhanced injection technique directly injects signals into the d - and q -axis currents and extracts the MTPA indicator from the reconstructed dc-link current. Hence, this enhanced injection technique avoids any transformations associated with polar coordinates, and it does not depend on a dc-link voltage sensor or a high-resolution speed sensor. The derived MTPA control law, unlike conventional MTPA control laws, can be realized in real time without mathematical approximations or lookup tables. With the combination of the enhanced signal injection technique and the derived MTPA control law, the proposed MTPA scheme can achieve an accurate and fast MTPA operation. Simulation and experimental results confirm the validity of the proposed MTPA scheme.

Index Terms—DC-link current, interior permanent magnet synchronous motor (IPMSM), maximum torque per ampere (MTPA), MTPA control law, parameter variation.

I. INTRODUCTION

INTERIOR permanent magnet synchronous motors (IPMSMs) are an excellent choice for electric drive systems because of their superior characteristics such as high efficiency, high power density, and wide speed range [1], [2]. Unlike a surface-mounted PMSM, the IPMSM contains a special arrangement of permanent magnets (PMs), providing opportunities to exploit not only the magnetic torque but also the reluctance torque [3], [4]. This implies that both the d - and q -axis stator currents can contribute to torque generation

of the IPMSM. According to the work of [5]–[8], different combinations of d - and q -axis currents for generating the desired torque can result in different motor losses. Therefore, a reasonable current control strategy should be adopted to improve the system efficiency.

Maximum torque per ampere (MTPA) control, which focuses on producing the required torque with the minimum current magnitude, is a frequently used efficiency optimization strategy for IPMSM drives. With this control strategy, the copper loss, which is the dominant loss of an IPMSM in the constant-torque region [9], can be minimized.

For a practical IPMSM, its parameters vary according to the operating conditions [10]–[13], posing a considerable challenge for MTPA control. To address this challenge, numerous robust MTPA schemes have been investigated. An experiment-based method was reported in [14], in which the MTPA points were obtained through a set of offline experiments. This method does not depend on any motor parameters but is highly time-consuming. As an alternative to performing offline tests, the MTPA points can also be acquired online by searching the current-vector-minimizing solution for a given operating condition [15]. In this way, the time-consuming offline experiments can be avoided. However, the search-based approach shows a slow dynamic response.

Recently, MTPA control methods based on real or virtual signal injection have been investigated [10], [16]–[20]. These methods have the advantage of robustness to motor parameter variations. In the injection-based MTPA methods, it is necessary to extract the MTPA indicator (i.e., the information that can indicate whether the MTPA control is achieved) from some system quantities. The patterns of the MTPA indicator extraction can be roughly classified into four categories:

- 1) MTPA indicator extraction from motor speed (or speed regulator output);
- 2) MTPA indicator extraction from electric power;
- 3) MTPA indicator extraction from virtual torque;
- 4) MTPA indicator extraction from stator flux linkage amplitude.

Unfortunately, the performance of the first extraction pattern is limited by the resolution of a position/speed sensor. The latter three extraction patterns are not sensitive to the resolution of the speed sensor. Nevertheless, these three extraction patterns depend on the stator voltages reconstructed from the dc-link

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voltage, and hence, they may not work properly when the dc-link voltage sensor fails or when no such sensor is present. Moreover, in previous injection-based methods, the injected signals for MTPA control are commonly based on the current vector angle in polar coordinates, introducing coordinate transformations associated with polar coordinates. To avoid these problems and meet the challenge of motor parameter variations, an enhanced signal injection technique for MTPA control is proposed in this paper.

In addition to the influence of parameter variations, dynamic behavior also needs to be focused in MTPA control. Compared with the search-based methods, the injection-based methods exhibit higher dynamic performance. However, the dynamic performance of the injection-based methods still needs to be enhanced because time is still required to converge to the MTPA points. An MTPA control law is beneficial to improve the dynamic performance of the injection-based methods, since it can directly calculate the constant-parameter model-based MTPA references. Among the MTPA control laws suitable for field-oriented control [21]–[24], the one in terms of the relationship between the d - and q -axis currents is simpler than the others [23]. However, the square-root computation, which is computationally intensive and processor-sensitive, is still inevitable. The typical solutions to this issue are lookup tables (LUTs) and low-order Taylor series approximations [25], [26]. Unfortunately, these practices can introduce new problems. More specifically, an LUT can result in a large amount of additional memory consumption, and low-order Taylor series approximations exhibit low accuracy at heavy loads. To avoid these problems and develop a fast MTPA control, a simple MTPA control law for calculating the constant-parameter MTPA references is derived in this paper.

In this study, a signal injection technique for MTPA control is proposed. This technique can cope with the influence of motor parameter variations on MTPA control accuracy. Unlike previous signal injection techniques, the proposed signal injection technique shows low dependence on system hardware and avoids the coordinate transformations associated with polar coordinates. Moreover, an MTPA control law is derived. The derived MTPA control law can enhance the dynamic performance of the injection-based MTPA control, and it does not involve any complicated computations that usually exist in conventional MTPA control laws. The overall MTPA control scheme combines the derived MTPA control law with the proposed signal injection technique, where the MTPA control law provides the constant-parameter model-based MTPA references, and the signal injection technique corrects the MTPA reference deviations caused by parameter variations. In this way, a fast and accurate MTPA operation can be achieved.

II. IPMSM MODEL AND MTPA INDICATOR

A. Mathematical Model of the IPMSM

When the major factors influencing the motor parameters are considered [10]–[13], the mathematical model of an IPMSM in

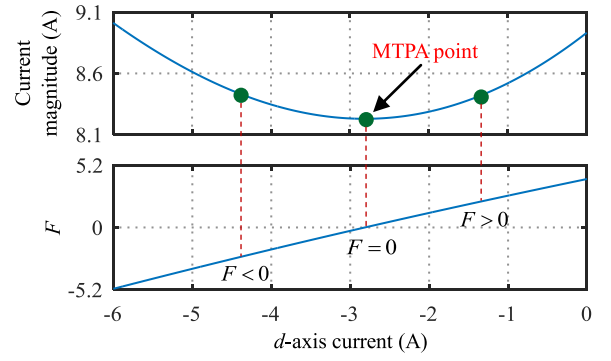


Fig. 1. Current magnitude and F versus d -axis current under a constant load torque.

the d - q reference frame can be expressed as

$$v_d = Ri_d + L_d(i_d, i_q) \frac{di_d}{dt} - p\omega_m L_q(i_d, i_q) i_q \quad (1)$$

$$v_q = Ri_q + L_q(i_d, i_q) \frac{di_q}{dt} + p\omega_m L_d(i_d, i_q) i_d + p\omega_m \psi_f(i_d, i_q) \quad (2)$$

$$T_e = \frac{3}{2} p \{ \psi_f(i_d, i_q) i_q + [L_d(i_d, i_q) - L_q(i_d, i_q)] i_d i_q \} \quad (3)$$

where $v_{d,q}$, $i_{d,q}$, and $L_{d,q}$ are the d - and q -axis voltages, currents, and inductances, respectively, and R , ω_m , ψ_f , T_e , and p are the stator resistance, mechanical angular speed, PM flux linkage, electromagnetic torque, and number of pole pairs, respectively.

B. MTPA Indicator

MTPA control implies generating the desired torque with the minimum stator current. Hence, the MTPA currents should satisfy the following constrained optimization problem:

$$\text{minimize } |i_s| = \sqrt{i_d^2 + i_q^2}, \quad \text{subject to } T_e = T_{e0} \quad (4)$$

where $|i_s|$ is the magnitude of the stator current, and T_{e0} is the desired torque. From (4), the MTPA condition can be derived as $i_d(\partial T_e / \partial i_q) - i_q(\partial T_e / \partial i_d) = 0$, and the corresponding derivation process is detailed in Appendix A. For convenience, let the left-hand side of this MTPA condition be denoted by F , i.e.,

$$F = i_d \frac{\partial T_e}{\partial i_q} - i_q \frac{\partial T_e}{\partial i_d}. \quad (5)$$

Then, the derived MTPA condition $i_d(\partial T_e / \partial i_q) - i_q(\partial T_e / \partial i_d) = 0$ can be summarized as $F = 0$. Fig. 1 shows the current magnitude and F under a constant load torque. As shown in Fig. 1, $F = 0$ is indeed satisfied at the MTPA point and the sign of F indicates that the operating point is located at the left or right side of the MTPA point. Also, the value of F reflects the distance between the operating point and the MTPA point. Therefore, F can serve as an MTPA indicator, which makes it possible to develop an accurate MTPA control.

III. PROPOSED SIGNAL INJECTION TECHNIQUE FOR MTPA CONTROL

To achieve an accurate MTPA operation that is not affected by motor parameter variations, a signal injection technique for MTPA control is presented in this section.

A. Torque Response Due to the Injected Signals

Usually, the injected signals are based on the current vector angle in polar coordinates, causing coordinate transformations associated with polar coordinates. To avoid this problem, the injected signals are directly injected into the d - and q -axis currents in this paper. The high-frequency (HF) signals injected into the d - and q -axis currents are designed as $-i_{q0}A\sin\omega_h t$ and $i_{d0}A\sin\omega_h t$, respectively, where i_{d0} and i_{q0} are the dc components of the d - and q -axis currents, respectively, ω_h is the injected frequency, and A ($0 < A < 0.08$) is the injection gain coefficient. Then, the d - and q -axis currents can be described as

$$\begin{cases} i_d = i_{d0} - i_{q0}A\sin\omega_h t \\ i_q = i_{q0} + i_{d0}A\sin\omega_h t \end{cases} \quad (6)$$

In this case, the Taylor series expansion of the resultant torque can be derived as

$$\begin{aligned} T_e(i_d, i_q) &= T_e(i_{d0} - i_{q0}A\sin\omega_h t, i_{q0} + i_{d0}A\sin\omega_h t) \\ &= T_e(i_{d0}, i_{q0}) + \left[i_{d0} \frac{\partial T_e}{\partial i_d}(i_{d0}, i_{q0}) \right. \\ &\quad \left. - i_{q0} \frac{\partial T_e}{\partial i_q}(i_{d0}, i_{q0}) \right] A\sin\omega_h t + \dots \end{aligned} \quad (7)$$

MTPA control is also called minimum copper loss control. As derived in Appendix B, the copper loss due to the injected components of i_d and i_q is significant smaller than the copper loss due to the original dc components of i_d and i_q . Hence, the injected HF signals themselves introduce almost no additional copper loss, and the minimum copper loss can be achieved by enabling the dc components of i_d and i_q follow the MTPA (i.e., minimum copper loss) current trajectory. In this case, it is reasonable to focus only on the dc component of the MTPA indicator F . From (5), the dc component of F is derived as

$$F(i_{d0}, i_{q0}) = i_{d0} \frac{\partial T_e}{\partial i_d}(i_{d0}, i_{q0}) - i_{q0} \frac{\partial T_e}{\partial i_q}(i_{d0}, i_{q0}). \quad (8)$$

By substituting (8) into (7), the torque can be rewritten as

$$\begin{aligned} T_e(i_d, i_q) &= T_e(i_{d0} - i_{q0}A\sin\omega_h t, i_{q0} + i_{d0}A\sin\omega_h t) \\ &= T_e(i_{d0}, i_{q0}) + F(i_{d0}, i_{q0}) \cdot A\sin\omega_h t + \dots \end{aligned} \quad (9)$$

As shown in (9), the torque contains information regarding $F(i_{d0}, i_{q0})$, which will be helpful in the extraction of $F(i_{d0}, i_{q0})$. Furthermore, it can be deduced from (9) that if $F(i_{d0}, i_{q0}) = 0$, the dominant pulsating component of the torque, which fluctuates at the injected frequency ω_h , can be eliminated. Therefore, the injected signals have a limited effect on the torque when the IPMSM is operating on (or near) the MTPA trajectory.

B. Relationship Between $F(i_{d0}, i_{q0})$ and DC-Link Current

Because the dc component of the MTPA indicator, namely, $F(i_{d0}, i_{q0})$, can be used to recognize whether the operating point (i_{d0}, i_{q0}) is the MTPA point, it is worth evaluating $F(i_{d0}, i_{q0})$. According to (9), $F(i_{d0}, i_{q0})$ can be extracted from the torque variation at the injected frequency ω_h . However, in practical applications, it is impracticable to measure the torque variation directly due to the absence of torque sensors. In this case, $F(i_{d0}, i_{q0})$ will be extracted from the dc-link current rather than the torque in this paper. In the following, the relationship between $F(i_{d0}, i_{q0})$ and the dc-link current will be detailed.

For the voltage source inverter (VSI), the dc-link current i_{dc} can be expressed as

$$i_{dc} = \frac{P_{in}}{v_{dc}} = \frac{P_{in0} + P_{in,h}}{v_{dc0} + v_{dc,h}} \quad (10)$$

where P_{in} is the input power of the VSI, P_{in0} and $P_{in,h}$ are the dc and HF components of P_{in} , respectively, v_{dc} is the dc-link voltage of the VSI, and v_{dc0} and $v_{dc,h}$ are the dc and HF components of v_{dc} , respectively. From (10), the bivariate Taylor series expansion of i_{dc} around (P_{in0}, v_{dc0}) is derived as

$$i_{dc} = \frac{P_{in0}}{v_{dc0}} + \frac{P_{in,h}}{v_{dc0}} - \frac{P_{in0}v_{dc,h}}{v_{dc0}^2} + \dots \quad (11)$$

Then, it is observed that specific information regarding $P_{in,h}$ and $v_{dc,h}$ is required to analyze the HF components in i_{dc} . Hence, further derivations of $P_{in,h}$ and $v_{dc,h}$ should be performed.

$P_{in,h}$ can be derived with the aid of P_{in} . Under the assumption of a lossless VSI, the input power of the VSI can be calculated as follows:

$$P_{in} = \frac{3}{2}(v_d i_d + v_q i_q). \quad (12)$$

Based on (1)–(3), (12) is expanded as follows:

$$\begin{aligned} P_{in} &= \frac{3}{2} \left[R(i_d^2 + i_q^2) + L_d(i_d, i_q) i_d \frac{di_d}{dt} + L_q(i_d, i_q) i_q \frac{di_q}{dt} \right] \\ &\quad + \omega_m T_e(i_d, i_q). \end{aligned} \quad (13)$$

Next, combining (6), (9), and (13) yields the input power considering the HF injection

$$\begin{aligned} P_{in} &= \frac{3}{2} R(i_{d0}^2 + i_{q0}^2) + \frac{3}{2} R(i_{d0}^2 + i_{q0}^2) A^2 \sin^2 \omega_h t \\ &\quad - \frac{3}{2} [L_d(i_d, i_q) - L_q(i_d, i_q)] i_{d0} i_{q0} A \omega_h \cos \omega_h t \\ &\quad + \frac{3}{4} [L_d(i_d, i_q) i_{q0}^2 + L_q(i_d, i_q) i_{d0}^2] A^2 \omega_h \sin 2\omega_h t \\ &\quad + \omega_m T_e(i_{d0}, i_{q0}) + \omega_m F(i_{d0}, i_{q0}) A \sin \omega_h t + \dots \end{aligned} \quad (14)$$

According to (14), the HF component of P_{in} (i.e., $P_{in,h}$) contains $\sin\omega_h t$, $\cos\omega_h t$, $\sin 2\omega_h t$, $\cos 2\omega_h t$, and high-order harmonic terms. This means that the HF component $P_{in,h}$ can simply be summarized as the Fourier series (15a), where P_{An} and P_{Bn} denote the amplitudes of the n th sine and cosine components of $P_{in,h}$, respectively. In the subsequent discussion,

$F(i_{d0}, i_{q0})$ is crucial, which is included in the amplitude of the $\sin\omega_h t$ term of (14). Hence, the amplitude of the $\sin\omega_h t$ term of (15a) (i.e., P_{A1}) should be focused accordingly. Then, based on the equivalence relation between the $\sin\omega_h t$ term of (14) and that of (15a), P_{A1} can be derived as $\omega_m F(i_{d0}, i_{q0})A$. In this case, (15a) can be rewritten as (15b)

$$P_{in,h} = P_{A1} \sin\omega_h t + P_{B1} \cos\omega_h t + \sum_{n=2}^{\infty} (P_{An} \sin n\omega_h t + P_{Bn} \cos n\omega_h t) \quad (15a)$$

$$P_{in,h} = \omega_m F(i_{d0}, i_{q0})A \sin\omega_h t + P_{B1} \cos\omega_h t + \sum_{n=2}^{\infty} (P_{An} \sin n\omega_h t + P_{Bn} \cos n\omega_h t). \quad (15b)$$

After the derivation of $P_{in,h}$, $v_{dc,h}$ also needs to be derived, since $v_{dc,h}$ is also a source of the HF components of i_{dc} , as shown in (11). One of the factors that may result in dc-link voltage fluctuation is the HF signal injection. Since the gain coefficient A of the injected signals is very small, the injected signals have very little effect on the dc-link voltage. In this case, the dc-link voltage fluctuation due to the injected signals can be neglected in the analysis. In addition to the injected signals, the other factor that may result in dc-link voltage fluctuation is the drive topology. In this paper, the VSI is fed by a three-phase diode rectifier. Due to the alternating turning-on and turning-off of the rectifier diodes, the fluctuating component $v_{dc,h}$ of the dc-link voltage primarily manifests as a signal at six times the grid frequency and the corresponding harmonics [27], i.e.,

$$v_{dc,h} = \sum_{n=1}^{\infty} (v_{An} \sin n\omega_{g6}t + v_{Bn} \cos n\omega_{g6}t) \quad (16)$$

where v_{An} and v_{Bn} denote the amplitudes of the n th sine and cosine components of the voltage, respectively, and ω_{g6} is six times the grid frequency.

After $P_{in,h}$ and $v_{dc,h}$ are derived, by substituting (15) and (16) into (11), a more detailed expression of i_{dc} (i.e., the relationship between i_{dc} and $F(i_{d0}, i_{q0})$) is obtained as

$$\begin{aligned} i_{dc} = & \frac{P_{in0}}{v_{dc0}} + \frac{A\omega_m F(i_{d0}, i_{q0})}{v_{dc0}} \sin\omega_h t + \frac{P_{B1}}{v_{dc0}} \cos\omega_h t \\ & + \frac{1}{v_{dc0}} \sum_{n=2}^{\infty} (P_{An} \sin n\omega_h t + P_{Bn} \cos n\omega_h t) \\ & - \frac{P_{in0}}{v_{dc0}^2} \sum_{n=1}^{\infty} (v_{An} \sin n\omega_{g6}t + v_{Bn} \cos n\omega_{g6}t) + \dots \end{aligned} \quad (17)$$

C. Extraction of $F(i_{d0}, i_{q0})$

Considering the bandwidth of the current loops, the injected frequency usually satisfies $0 < \omega_h < 2\omega_{g6}$. Under these constraints, it can be seen from (17) that if $\omega_h \neq \omega_{g6}$, the magnitude of the $\sin\omega_h t$ component is proportional to the dc component $F(i_{d0}, i_{q0})$ of the MTPA indicator. Then, the information re-

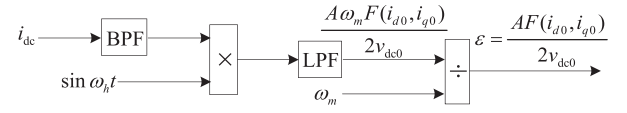


Fig. 2. Demodulation operation for extracting the information of $F(i_{d0}, i_{q0})$.

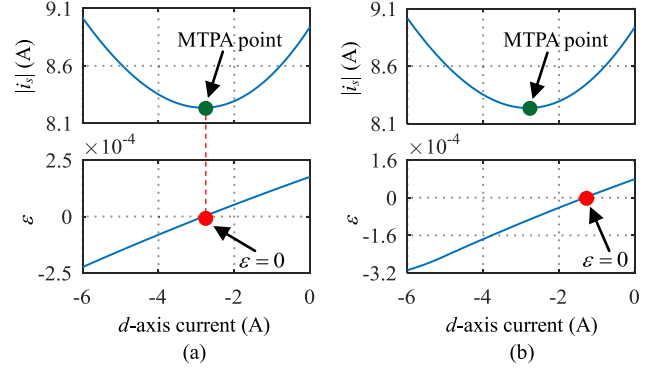


Fig. 3. Current magnitude and ε (extracted from i_{dc}) versus d -axis current under a constant load torque. (a) $\omega_h \neq \omega_{g6}$. (b) $\omega_h = \omega_{g6}$.

garding $F(i_{d0}, i_{q0})$ can be extracted from i_{dc} by means of a demodulation operation, as shown in Fig. 2.

In Fig. 2, the dc-link current is filtered by a bandpass filter (BPF) whose center frequency is ω_h , such that only the injected frequency component [$\sin\omega_h t$ and $\cos\omega_h t$ terms in (17)] is obtained. The output of the BPF is then multiplied by $\sin\omega_h t$, and the product can be described as

$$\begin{aligned} & \left[\frac{A\omega_m F(i_{d0}, i_{q0})}{v_{dc0}} \sin\omega_h t + \frac{P_{B1}}{v_{dc0}} \cos\omega_h t \right] \sin\omega_h t \\ & = \frac{A\omega_m F(i_{d0}, i_{q0})}{v_{dc0}} \sin^2\omega_h t + \frac{P_{B1}}{v_{dc0}} \cos\omega_h t \cdot \sin\omega_h t \\ & = \frac{A\omega_m F(i_{d0}, i_{q0})}{2v_{dc0}} - \frac{A\omega_m F(i_{d0}, i_{q0})}{2v_{dc0}} \cos 2\omega_h t \\ & \quad + \frac{P_{B1}}{2v_{dc0}} \sin 2\omega_h t. \end{aligned} \quad (18)$$

To obtain only the dc component, this product is fed into a low-pass filter (LPF). Finally, dividing the obtained dc component $A\omega_m F(i_{d0}, i_{q0})/(2v_{dc0})$ by ω_m yields the desired information regarding $F(i_{d0}, i_{q0})$, which can be represented as

$$\varepsilon = \frac{AF(i_{d0}, i_{q0})}{2v_{dc0}}. \quad (19)$$

This implies that the information extracted from i_{dc} is proportional to the dc component $F(i_{d0}, i_{q0})$ of the MTPA indicator. Benefitting from this proportional relationship, ε can also indicate the relative location of the operating point with respect to the MTPA point, and the relationship $\varepsilon = 0$ is satisfied at the MTPA point. These characteristics of ε are well illustrated by the simulation results presented in Fig. 3(a).

When ω_h is designed as $\omega_h = \omega_{g6}$, (17) can be rewritten as

$$\begin{aligned} i_{dc} = & \frac{P_{in0}}{v_{dc0}} + \left[\frac{A\omega_m F(i_{d0}, i_{q0})}{v_{dc0}} - \frac{P_{in0}v_{A1}}{v_{dc0}^2} \right] \sin \omega_h t \\ & + \left(\frac{P_{B1}}{v_{dc0}} - \frac{P_{in0}v_{B1}}{v_{dc0}^2} \right) \cos \omega_h t \\ & + \sum_{n=2}^{\infty} \left(\frac{P_{An}}{v_{dc0}} - \frac{P_{in0}v_{An}}{v_{dc0}^2} \right) \sin n\omega_h t \\ & + \sum_{n=2}^{\infty} \left(\frac{P_{Bn}}{v_{dc0}} - \frac{P_{in0}v_{Bn}}{v_{dc0}^2} \right) \cos n\omega_h t + \dots \quad (20) \end{aligned}$$

Under this condition, the information extracted from i_{dc} using the demodulation operation illustrated in Fig. 2 becomes

$$\varepsilon = \frac{AF(i_{d0}, i_{q0})}{2v_{dc0}} - \frac{P_{in0}v_{A1}}{2\omega_m v_{dc0}^2}. \quad (21)$$

Then, the information extracted from i_{dc} is no longer proportional to $F(i_{d0}, i_{q0})$. Thus, in the case of $\omega_h = \omega_{g6}$, the operating point satisfying $\varepsilon = 0$ may not be the MTPA point, as shown in Fig. 3(b). By substituting $F(i_{d0}, i_{q0}) = 0$ into (21), it can be deduced that in the case of $\omega_h = \omega_{g6}$, the MTPA point satisfies $\varepsilon = -P_{in0}v_{A1}/(2\omega_m v_{dc0}^2)$. Because v_{A1} is unknown when the dc-link voltage sensor fails or when no such sensor is present, ε cannot be used to determine whether the MTPA point is successfully tracked in the case of $\omega_h = \omega_{g6}$. To avoid this problem, ω_h should be chosen as a frequency that is not near ω_{g6} . In this paper, ω_{g6} is 600π rad/s and the injected frequency ω_h is 320π rad/s.

Moreover, to extract the information regarding $F(i_{d0}, i_{q0})$ using the demodulation operation shown in Fig. 2, the relevant filters need to be designed. The input of the LPF can be expressed by (18), which contains both the dc component and the ac component ($\sin 2\omega_h t$ and $\cos 2\omega_h t$). To obtain only the dc component of this product, the cutoff frequency of the LPF should be much less than $2\omega_h$, preferably less than $0.3\omega_h$. For the BPF, the bandwidth needs to be selected in addition to the center frequency. When the bandwidth of the BPF is smaller than its center frequency, the ripple of the demodulation output increases with the bandwidth of the BPF, and the dynamic performance of the demodulation decreases with the bandwidth of the BPF. Considering these characteristics, it is better to select the bandwidth of the BPF within the range $0.01\omega_h \sim 0.3\omega_h$.

D. Reconstruction of the DC-Link Current

Since i_{dc} is the source for extracting the information regarding $F(i_{d0}, i_{q0})$, it is necessary to explain the method of obtaining i_{dc} . Equations (17) and (20) are provided for indicating that the information regarding $F(i_{d0}, i_{q0})$ is contained in i_{dc} , which are not suitable for calculating i_{dc} . In practice, i_{dc} can be easily obtained without a dc-link current sensor because it can be reconstructed using the motor phase currents as follows:

$$\hat{i}_{dc} = (T_a i_a + T_b i_b + T_c i_c)/T_s \quad (22)$$

where $\hat{i}_{dc}$ denotes the reconstructed i_{dc} , T_a , T_b , and T_c are the upper-switch gating time of each VSI leg, i_a , i_b , and i_c are

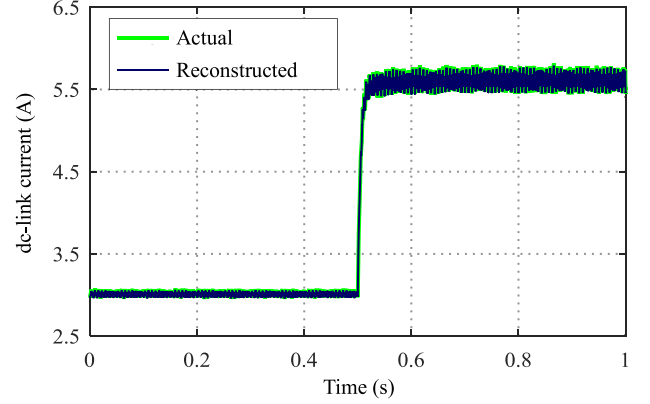


Fig. 4. Reconstructed and actual dc-link currents.

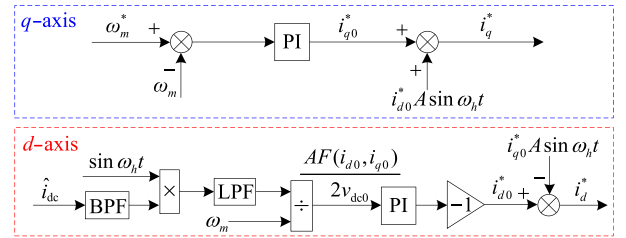


Fig. 5. Block diagram of the proposed signal injection technique.

the motor phase currents, and T_s is the switching period. To demonstrate the validity of (22), Fig. 4 shows both the reconstructed and actual dc-link currents. It can be seen that satisfactory reconstruction accuracy can be achieved using (22), either at steady and transient states. It should be noted that in Fig. 4, the switching frequency components of the actual dc-link current are filtered out for convenient observation.

E. Configuration of the Proposed Signal Injection Technique

The proposed signal injection technique is shown in Fig. 5, where the superscript “*” denotes the reference. The reference for the dc component of the q -axis current, namely, i_{q0}^* , is provided by the speed proportional-integral (PI) regulator, and the reference for the dc component of the d -axis current, namely, i_{d0}^* , is regulated according to the information extracted from $\hat{i}_{dc}$. If the operating point deviates from the MTPA point, the information regarding $F(i_{d0}, i_{q0})$ extracted from $\hat{i}_{dc}$, namely, $AF(i_{d0}, i_{q0})/(2v_{dc0})$, will be nonzero. Then, i_{d0}^* will be modified by the PI regulator until the quantity $AF(i_{d0}, i_{q0})/(2v_{dc0})$, which is proportional to $F(i_{d0}, i_{q0})$, fluctuates around zero. In other words, the deviation between the operating point and the MTPA point can be eliminated.

The HF signals in previous studies are commonly injected into the current angle in polar coordinates, whereas the HF signals in this study are directly injected into the d - and q -axis currents. Hence, the proposed signal injection technique avoids the coordinate transformations associated with polar coordinates. In addition, the previous injection methods extract the MTPA

indicator information from motor speed or the system quantities calculated using dc-link voltage, whereas the proposed technique extracts the MTPA indicator information from the dc-link current reconstructed using motor phase currents. Thus, unlike the existing signal injection techniques, the presented technique does not depend on a high-resolution speed sensor or a dc-link voltage sensor. That is, the presented technique exhibits low dependence on system hardware.

IV. CONSTANT-PARAMETER MTPA CONTROL LAW

When the load torque changes rapidly, it takes time for the injection-based MTPA methods to converge to the MTPA points. For the proposed signal injection technique to have a high rate of convergence during load variations, the MTPA control law, which can directly calculate the constant-parameter model-based MTPA references, is a good assist means. Assuming that the motor parameters are constant, the stator currents for the MTPA condition can be derived as [23]

$$i_d = \frac{\psi_f}{2(L_q - L_d)} - \sqrt{\frac{\psi_f^2}{4(L_q - L_d)^2} + i_q^2}. \quad (23)$$

This MTPA control law is somewhat complex (particularly for low-performance microcontrollers) because of the square-root operation, and thus, an LUT or a Taylor series approximation is usually employed for real-time implementation [25], [26]. However, such a practice may lead to additional memory consumption or large current reference deviations. Therefore, a simple MTPA control law is required.

From [28], the MTPA control law (23) can be rewritten as

$$(\Lambda - i_d)^2 = \Lambda^2 + i_q^2 \quad (24)$$

where

$$\Lambda = \psi_f / [2(L_q - L_d)] \quad (25)$$

is a combined motor parameter containing ψ_f , L_d , and L_q . From (24), the MTPA condition can be further rewritten as

$$i_q = 2\Lambda(-i_d/i_q) / [1 - (i_d/i_q)^2]. \quad (26)$$

Let i_K denotes the $(-i_d/i_q)$ terms of (26), i.e.,

$$i_K = -i_d/i_q, \quad i_{d,q} \text{ subject to (26)}. \quad (27)$$

Then, according to (26) and (27), the MTPA i_d and i_q can be expressed in terms of i_K , i.e.,

$$\begin{cases} i_q = 2\Lambda i_K / (1 - i_K^2) \\ i_d = -i_q \cdot i_K \end{cases} \quad (28)$$

For a given i_K , the corresponding MTPA i_d and i_q can be calculated uniquely using (28). Compared with the conventional constant-parameter MTPA control laws, the derived MTPA control law (28) does not contain any complicated operations and thus can be calculated in real time without LUTs or Taylor series approximations.

V. PROPOSED OVERALL MTPA CONTROL SCHEME

To achieve an accurate and fast MTPA operation, the overall MTPA control scheme is based on the proposed signal injection

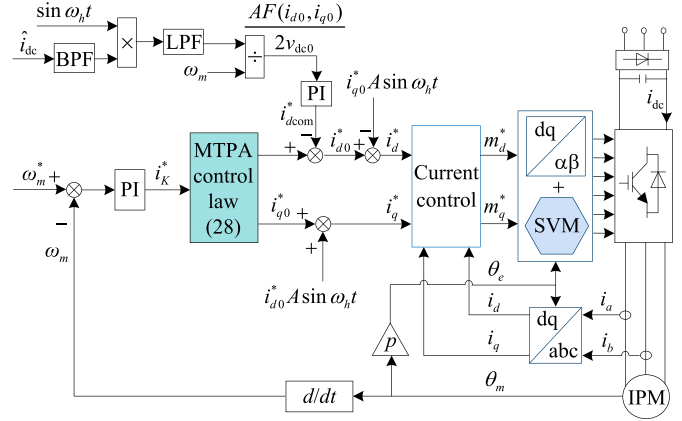


Fig. 6. Block diagram of the proposed overall MTPA control scheme.

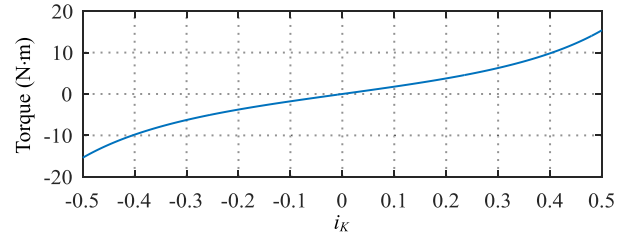


Fig. 7. Relationship between torque and i_K under constant-parameter MTPA condition.

technique and the derived MTPA control law, as shown in Fig. 6. It should be noted that in Fig. 6, θ_e is the electrical angle.

Substituting (28) into the constant-parameter torque equation $T_e = \frac{3}{2}p[\psi_f i_q + (L_d - L_q)i_d i_q]$, the motor torque under constant-parameter MTPA condition can be expressed in terms of i_K as

$$T_e = 3\Lambda p \psi_f i_K / (i_K^2 - 1)^2. \quad (29)$$

In this case, i_K is positively correlated with the motor torque, as depicted in Fig. 7. This implies that i_K can replace the torque to regulate the motor speed ω_m . Thus, in the overall MTPA scheme, the output of the speed PI regulator is configured as the command i_K^* . Once i_K^* is given by the speed regulator, the constant-parameter model-based MTPA current references can be calculated directly with the MTPA control law (28), which contributes to achieving a fast MTPA operation. In the overall MTPA scheme, since the MTPA control law (28) is responsible for enhancing the dynamic performance of MTPA control, it only needs to provide the dominant components of MTPA references. In this case, for simplicity, it is reasonable to assume that the motor parameters are constant during the derivation of (28).

In the overall MTPA scheme shown in Fig. 6, the signal injection technique is responsible for addressing the effect of parameter variations on the MTPA control accuracy. Under the excitation of the currents in the form of (6), the information regarding $F(i_{d0}, i_{q0})$ can be extracted from the reconstructed dc-link current $\hat{i}_{dc}$, as discussed in Section III. Because the quantity $AF(i_{d0}, i_{q0})/(2v_{dc0})$ extracted from $\hat{i}_{dc}$ reflects how far the

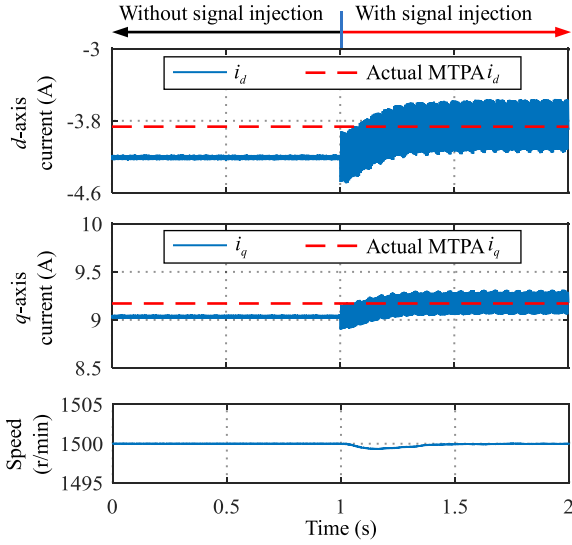


Fig. 8. Simulation results with and without the signal injection technique being enabled under 11 N·m at 1500 r/min.

operating point is from the MTPA point, this quantity is used to compensate for the d -axis current reference deviation (between the actual MTPA d -axis current reference and the d -axis current reference provided by the MTPA control law) through a PI regulator. With the adjustment of this PI regulator, the quantity $AF(i_{d0}, i_{q0})/(2v_{dc0})$ extracted from $\hat{i}_{dc}$ converges to zero, and the deviation between the operating point and the MTPA point due to parameter variations can be eliminated accordingly. Thus, an accurate MTPA control can be achieved.

To demonstrate the compensation effect of the signal injection technique, Fig. 8 shows the simulation results with and without the signal injection technique being enabled under 11 N·m at 1500 r/min. Initially, the signal injection technique was disabled, and the current references were only provided by the constant-parameter MTPA control law (28). In this case, the d - and q -axis currents slightly deviated from the actual MTPA currents. After the signal injection technique was enabled, the d - and q -axis currents were adjusted automatically until the deviation between the operating point and the actual MTPA point was corrected successfully. During the correction transient, the d -axis current correction signal i_{dcom}^* (see Fig. 6) may cause torque variation. However, since the d -axis current is not the dominant current component that can generate motor torque, and the value of the correction signal i_{dcom}^* is relatively small, the torque variation due to the correction transient can be easily eliminated by the speed controller. In this case, the correction transient of the d -axis current has very limited effect on the motor speed, as shown in Fig. 8.

VI. SIMULATION VERIFICATION

Simulations were performed using MATLAB/SIMULINK to validate the proposed overall MTPA control scheme. The dc-link of the drive system is established using a three-phase diode rectifier whose input line voltage and smoothing capacitance are 248 V and 1000 μ F, respectively. The IPMSM model is

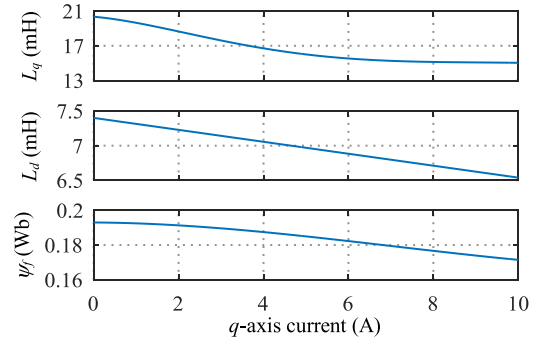


Fig. 9. Parameters of the IPMSM model.

TABLE I
KEY PARAMETERS OF THE IPMSM

Rated power	2.3 kW	Pole pairs	4
Rated torque	11.0 N·m	Stator resistance	1.0 Ω
Rated current	7.1 A (rms)	PM flux linkage (no load)	0.193 Wb
Rated speed	2000 r/min	d - q axis inductances (no load)	7.4/20.1 mH

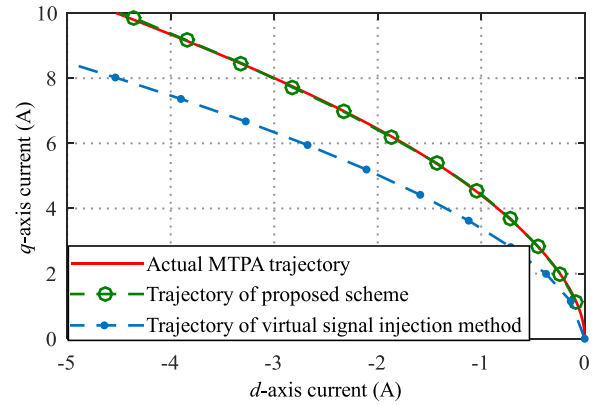


Fig. 10. Simulation results of the proposed MTPA scheme and the virtual signal injection method.

parameter variable (refer to Fig. 9) and its key parameters are listed in Table I. The preceding results in Figs. 1, 3, 4, and 8 are all based on this IPMSM model.

The steady-state performance of the proposed scheme was simulated under different load torques, and the resultant current trajectory is depicted in Fig. 10. Additionally, the actual MTPA trajectory and the current trajectory of the virtual signal injection method [10] are also shown in Fig. 10, where the actual MTPA trajectory is connected by the current-minimizing solutions under different load torques. It can be observed that the current trajectory of the proposed scheme matches well with the actual MTPA trajectory, and its MTPA tracking accuracy is higher than that of the virtual signal injection method.

Since MTPA control can be considered as minimum copper loss control, the control accuracy comparison between the proposed scheme and the virtual signal injection method can also be conducted from the copper loss perspective. Table II lists the copper losses of these two methods under various load levels. As can be seen, the proposed scheme consumes lower copper loss

TABLE II
COMPARISON OF COPPER LOSS UNDER DIFFERENT LOAD TORQUES

Load torque (N·m)	Copper loss of proposed scheme (W)	Copper loss of virtual signal injection method (W)	Difference (%)
1.0	2.01	2.01	0
3.0	12.56	12.66	0.80
5.0	32.60	33.08	1.47
7.0	62.58	63.82	1.98
9.0	101.80	104.15	2.31
11.0	148.43	152.25	2.57
12.0	174.05	178.61	2.62

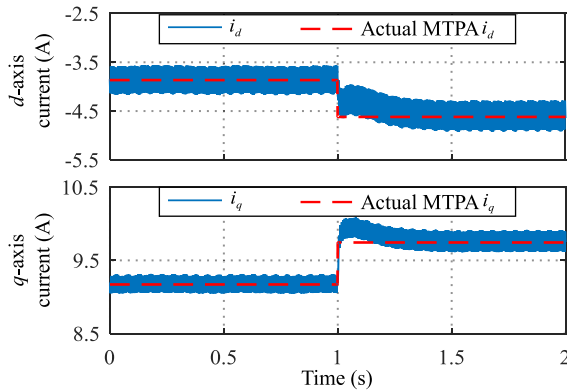


Fig. 11. Performance of the proposed MTPA scheme under 11 N·m when temperature changes.

than the virtual signal injection method, especially under heavy loads. This implies that the proposed scheme is much closer to the ideal MTPA control than the virtual signal injection method.

The PM flux linkage may vary with the temperature, and the influence of this characteristic on the proposed scheme was investigated by simulation under a load torque of 11 N·m. To represent the variation of the PM flux linkage due to temperature rise, the PM flux linkage was reduced to 90% of its original value at time $t = 1$ s, and the d - and q -axis currents are shown in Fig. 11. It can be seen that although the actual MTPA d - and q -axis currents were changed (at $t = 1$ s) due to the variation of the PM flux linkage, the new actual MTPA d - and q -axis currents can be tracked by the proposed scheme. Hence, the proposed scheme is robust to motor parameter variations.

The transient response of the proposed MTPA scheme was simulated at 1200 r/min under a step load torque of 6–11–6 N·m, and the results are depicted in Fig. 12. It can be observed that the ripple frequency of the d - and q -axis currents is 160 Hz, which is the same as the injected frequency. Moreover, it can be concluded that the proposed MTPA scheme is robust to load disturbances and shows a fast MTPA tracking response.

VII. EXPERIMENTAL RESULTS

The proposed overall MTPA control scheme shown in Fig. 6 was evaluated experimentally on an IPMSM drive system. The experimental setup is shown in Fig. 13. During the tests, the IPMSM was controlled in speed control mode, and the load was

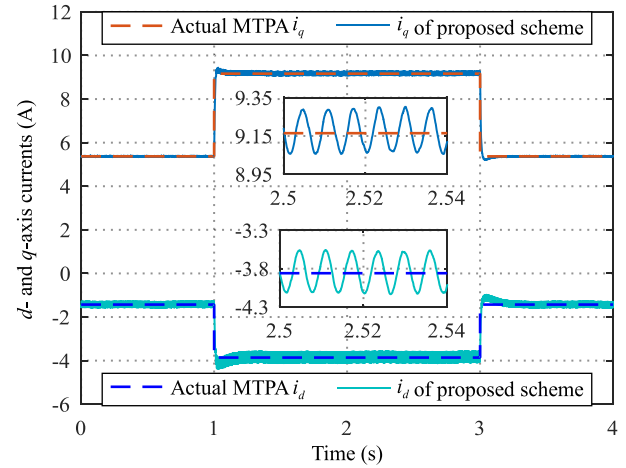


Fig. 12. Simulation dynamic response of the proposed MTPA scheme at 1200 r/min under a step load torque of 6–11–6 N·m.

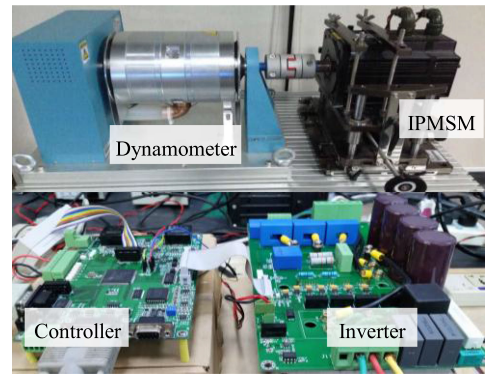


Fig. 13. Experimental setup.

provided by the dynamometer connected with the IPMSM. The relevant parameters of the IPMSM are presented in Table I, and the switching frequency is 10 kHz.

In IPMSM drives, the purpose of replacing conventional $i_d = 0$ strategy with MTPA strategy is to reduce the current magnitude. Therefore, a current magnitude comparison between the $i_d = 0$ method and the proposed MTPA scheme was performed under a load torque of 10 N·m, and the results are depicted in Fig. 14. Fig. 14 shows that the d -axis current changed from zero to negative values when the control algorithm was switched from the $i_d = 0$ method to the proposed MTPA scheme, and the current magnitude was decreased accordingly. In other words, the proposed MTPA scheme exhibits lower current magnitude than the $i_d = 0$ method under the given load torque.

Also, the current magnitudes of the proposed MTPA scheme and the conventional $i_d = 0$ method were compared under other load torques, and the results are summarized in Fig. 15. It can be observed that, for any given load torque, the current magnitude of the proposed MTPA scheme is lower than that of the $i_d = 0$ method. Thus, the proposed MTPA scheme can reduce the stator current magnitude under different load torques.

The phase currents at 1500 r/min under two constant load torques were measured and are shown in Fig. 16. It can be seen

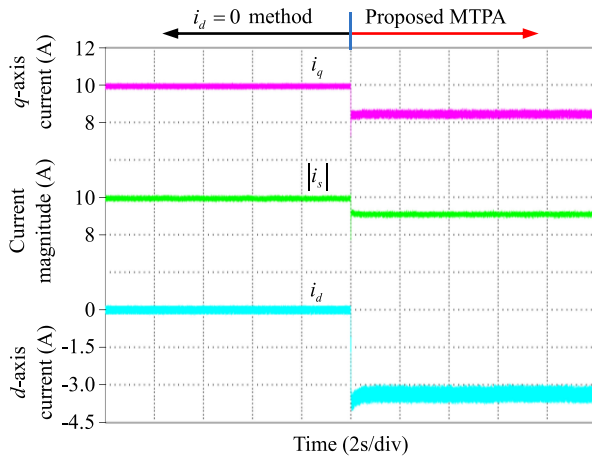


Fig. 14. Comparison between the $i_d = 0$ method and the proposed MTPA scheme under a load torque of 10 N·m.

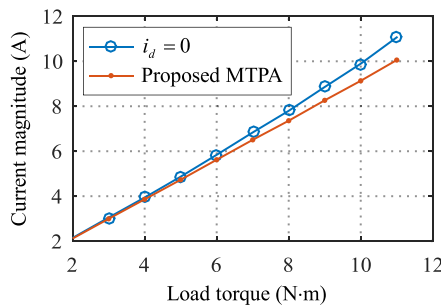


Fig. 15. Current magnitude comparison between the proposed MTPA scheme and the $i_d = 0$ method under different load torques.

that the phase currents approximate to pure sine waves, which implies that the injected signals have little effect on the motor phase currents due to the small gain coefficient ($A = 0.03$) of the injected signals.

To investigate the MTPA control accuracy of the proposed scheme, the IPMSM was operated under various load torques, and the current trajectory of the proposed scheme connected by the steady-state operating points is shown in Fig. 17. For comparison, the actual MTPA trajectory obtained by offline testing and the current trajectory of the virtual signal injection method [10] are also shown in Fig. 17. It can be seen that the virtual signal injection method generates a current trajectory that deviates from the actual MTPA trajectory. This is because the virtual signal injection method does not fully consider the motor parameter variations, as discussed in [29]. Unlike the virtual signal injection method, the proposed scheme generates a current trajectory that agrees well with the actual MTPA trajectory. Thus, the proposed scheme exhibits a higher MTPA control accuracy than the virtual signal injection method.

To study the influence of the dead time on the accuracy of the proposed scheme, the dead time was specially increased to 5 μ s without compensation. The current trajectory of the proposed scheme is shown in Fig. 18. From Fig. 18, the difference between the current trajectory of the proposed scheme and the

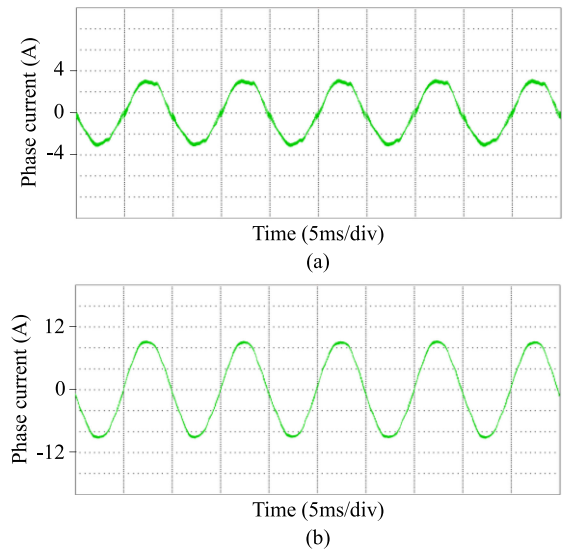


Fig. 16. Measured phase currents at 1500 r/min under constant load torques. (a) 3 N·m. (b) 10 N·m.

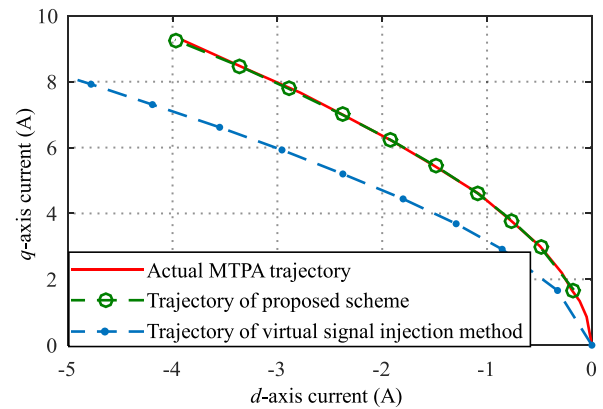


Fig. 17. Experimental results of the proposed MTPA scheme and the virtual signal injection method.

actual MTPA trajectory is very small. Therefore, the dead time usually has little effect on the accuracy of the proposed scheme. It should be noted that there is a special case, i.e., the frequency of the injected signals is close to that of the d - and q -axis voltage/current harmonics caused by the dead time. In this special case, the influence of the dead time on the MTPA control accuracy will be increased, and this influence can be lowered by changing the frequency of the injected signals or enabling the dead time compensation.

In addition, an experiment was performed at 700 r/min with a step load disturbance 7–10–7 N·m to demonstrate the dynamic behavior of the proposed MTPA control scheme. The experimental current and speed responses are shown in Fig. 19. As can be observed, when the load torque changed abruptly, the d - and q -axis currents were adjusted automatically until a new MTPA condition was reached. Hence, the capability of the proposed MTPA scheme to tolerate load disturbances is confirmed experimentally. Fig. 19 also demonstrates that the proposed scheme can achieve a satisfactory dynamic response.

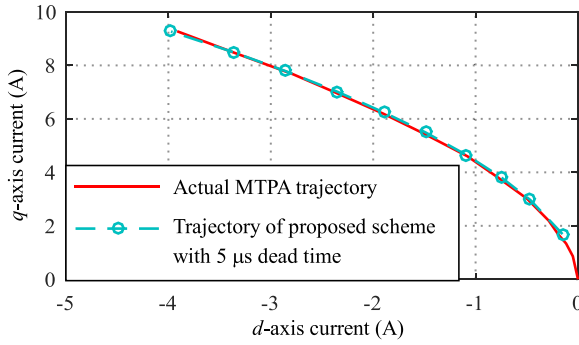


Fig. 18. Experimental current trajectory of the proposed MTPA scheme when the dead time is 5 μ s without compensation.

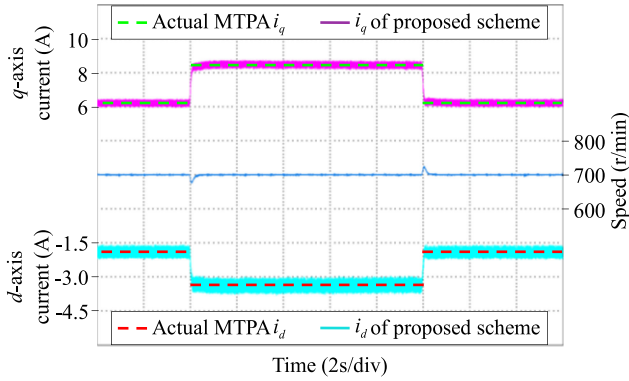


Fig. 19. Experimental dynamic response of the proposed MTPA scheme at 700 r/min with a step load disturbance 7–10–7 N·m.

VIII. DISCUSSION

MTPA control, which can minimize the copper loss, is a typical current control strategy below the base speed (i.e., in the constant-torque region), and hence, this study focuses on the region below the base speed. Within this region, the iron loss is much smaller than the copper loss [9], and its effect on the MTPA points is very limited [10]. Therefore, the iron loss is not considered in this study. When the IPMSM operates beyond the base speed (i.e., in the constant-power region), the iron loss will increase obviously. In this case, the maximum efficiency control will be more appropriate for reducing the motor loss.

IX. CONCLUSION

A signal injection technique has been proposed to cope with the influence of motor parameter variations on MTPA control. This technique utilizes the fact that the information on the MTPA indicator can be extracted from the dc-link current reconstructed using phase currents. In contrast to previous injection methods, the proposed signal injection technique avoids the transformations associated with polar coordinates and does not depend on a dc-link voltage sensor or a high-resolution speed sensor. In addition, to improve the dynamic performance of the MTPA operation, a simple MTPA control law has been derived. With this MTPA control law, the MTPA current references based on the

constant-parameter model can be calculated in real time without Taylor series approximations or LUTs. The proposed overall MTPA control scheme combines the derived MTPA control law with the proposed signal injection technique. The simulation and experimental results demonstrate that the proposed MTPA scheme can achieve an MTPA operation fast and accurately.

APPENDIX A

The Lagrange multiplier method is a good choice for solving the constrained optimization problem (4), and the Lagrange function for this optimization problem can be constructed as

$$L(i_d, i_q, \lambda) = \sqrt{i_d^2 + i_q^2} + \lambda(T_e - T_{e0}) \quad (A1)$$

where λ is the Lagrange multiplier. According to the necessary conditions for the optimality, the MTPA currents should satisfy

$$\begin{cases} \frac{\partial L}{\partial i_d} = \frac{i_d}{\sqrt{i_d^2 + i_q^2}} + \lambda \frac{\partial T_e}{\partial i_d} = 0 \\ \frac{\partial L}{\partial i_q} = \frac{i_q}{\sqrt{i_d^2 + i_q^2}} + \lambda \frac{\partial T_e}{\partial i_q} = 0 \\ \frac{\partial L}{\partial \lambda} = T_e - T_{e0} = 0 \end{cases} \quad (A2)$$

Then, the MTPA condition $i_d(\partial T_e / \partial i_q) - i_q(\partial T_e / \partial i_d) = 0$ can be derived from (A2).

APPENDIX B

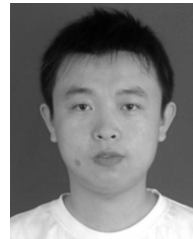
From (6), the average copper loss of the IPMSM over an injected period T_h ($T_h = 2\pi/\omega_h$) can be expressed as

$$\begin{aligned} \bar{P}_{cu} &= (1/T_h) \cdot \int_0^{T_h} \frac{3}{2} R(i_d^2 + i_q^2) dt \\ &= (1/T_h) \cdot \int_0^{T_h} \frac{3}{2} R \left[(i_{d0} - i_{q0} A \sin \omega_h t)^2 \right. \\ &\quad \left. + (i_{q0} + i_{d0} A \sin \omega_h t)^2 \right] dt \\ &= \frac{3}{2} R(i_{d0}^2 + i_{q0}^2) + \frac{3}{2} R(i_{d0}^2 + i_{q0}^2)(A^2/2). \end{aligned} \quad (A3)$$

Since i_d and i_q contain the original dc components and the injected HF components [see (6)], the copper loss can be divided into two parts: 1) the copper loss due to the dc components of i_d and i_q ; 2) the copper loss due to the injected components of i_d and i_q . Substituting the injection coefficient $A = 0$ into (A3), the copper loss due to the dc components of i_d and i_q can be obtained as $\frac{3}{2} R(i_{d0}^2 + i_{q0}^2)$. Then, subtracting $\frac{3}{2} R(i_{d0}^2 + i_{q0}^2)$ from (A3), the copper loss due to the injected components of i_d and i_q is obtained as $\frac{3}{2} R(i_{d0}^2 + i_{q0}^2)(A^2/2)$. In this paper, A is set to 0.03, and hence, the copper loss due to the injected components is $\frac{3}{2} R(i_{d0}^2 + i_{q0}^2) \cdot (0.00045)$. This means that, for the same i_{d0} and i_{q0} , the copper loss due to the injected signals themselves is only 0.00045 times the copper loss due to the original dc components of i_d and i_q . Hence, the injected signals themselves introduce almost no additional copper loss.

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